

Grader Output

C1M4_Programming

100 / 100 points earned

9 / 9 autograded cells passed

Graded Cells

Cell 4 (cell-ff0306c936b2973)

Passed | 18.87 / 18.87 points
View feedback

Cell 6 (cell-8db8b258b550abec)

Passed | 13.21 / 13.21 points
View feedback

Cell 8 (cell-7149dbcf2ac7fd20)

Passed | 7.55 / 7.55 points
View feedback

Cell 10 (cell-20bf85b6ed9e8fa2)

Passed | 9.43 / 9.43 points
View feedback

Cell 12 (cell-0d873558be79b01c)

Passed | 1.89 / 1.89 points
View feedback

Cell 14 (cell-6563da9a412bbf97)

Passed | 5.66 / 5.66 points
View feedback

Cell 16 (cell-c660f1c0ee6b1c03)

Passed | 18.87 / 18.87 points
View feedback

Cell 18 (cell-9706082123fc63cd)

Passed | 18.87 / 18.87 points
View feedback

Cell 20 (cell-8d731d6bde318a0f)

Passed | 5.66 / 5.66 points
View feedback

Module 4: Programming Assignment¶

```
In [1]: library(testthat)
library(ggplot2)

expect_constant = function(actual, tolerance = .Machine$double.eps*0.5) {
  if (length(actual) == 0) {
    fail("The vector is empty.")
  }

  # Check if all elements are the same within the given tolerance
  if (all(abs(actual - actual[1]) < tolerance)) {
    succeed()
  } else {
    fail("Your answer is not proportional to the posterior.")
  }
}
```

Problem 1: An objective analysis in R¶

1 (a) [10 points] Assume that the data $\mathbf{x} = (0.4478, 0.6173, 1.1317, 0.9011, 1.9250)$ are realizations from $X_1, X_2, X_3, X_4, X_5 \stackrel{iid}{\sim} \Gamma(1, \lambda)$. Conduct an "objective" Bayesian analysis using Jeffreys prior, $\pi(\lambda)$, and data $\mathbf{x}$.

Note: In videos in this module, we saw that, for parameter λ , $\pi(\lambda) \propto \sqrt{I(\lambda)}$, where $I(\lambda)$ is the Fisher Information number for the data. For the purposes of this problem use $\pi(\lambda) = 1/20\sqrt{I(\lambda)}$ when calculating the prior, for reasons that will become clear later in the question.

Step by step:

- Start by mathematically deriving the Jeffreys prior in this context. HINT: from a video in this module, the Fisher Information is most easily computed using the formula

$$I(\lambda) = -E\left[\frac{d^2}{d\lambda^2}\log(f(\mathbf{x}|\lambda))\right],$$

where $f(\mathbf{x}|\lambda)$ is the likelihood function for the data. Be sure to multiply by $c = 1/20$ to obtain $\pi(\lambda)$.

- Then, in R, compute this Jeffreys prior for $m = 200$ evenly spaced λ values between 0.1 and 2 (i.e., use `lambda = seq(0.1,2,length.out=200)`). Store the results in a vector labeled, `prior`.
- Finally, in R, compute exact posterior distribution (including the denominator of Bayes Theorem!) given the data for the same $m = 200$ evenly spaced λ values between 0.1 and 2. Store the values for the posterior in a vector labeled, `post`.

Use R functions for your answer; this will make part (c) easier!

In [2]: `x = c(0.4478, 0.6173, 1.1317, 0.9011, 1.9250)`

In [3]: Student's answer (Top)

```
# your code here
#.NotYetImplemented()
n = length(x)
#mean(x)
lambda = seq(0.1,2,length.out=200)
#prior = (1/20) * (sqrt(n) / lambda)

prior_func <- function(lambda) {
  (1/20) * (sqrt(n) / lambda)
}

likelihood_func <- function(lambda) {
  ll = 1
  for (i in 1:n) {
    ll <- ll * dgamma(x[i], shape=1, rate=lambda)
  }
  ll
}

posterior_func <- function(lambda) {
  likelihood_func(lambda)*prior_func(lambda) / integrate(function(l) likelihood_func(l)*prior_func(l), 0, Inf)$val
}

prior <- prior_func(lambda)
post <- posterior_func(lambda)

#L <- sapply(lambda, likelihood)
#L <- lambda^n * exp(-lambda * sum(x))
#post <- L*prior / sum(L*prior)

#post <- dgamma(lambda, shape=n, rate=sum(x))
#post
```

In [4]: Grade cell: cell-ff0306c936b2973 Score: 18.87 / 18.87 (Top)

Hidden Tests Redacted

Congratulations! All test cases in this cell passed.

1 (b) [7 points] Plot the prior and posterior distributions on the same plot using ggplot and store the plot as variable, `p`.

(The reason we chose a prior with a multiplier of $c = 1/20$ was so that the prior and posterior would be scaled for the same plot!)

In [5]: Student's answer (Top)

```
# your code here
#.NotYetImplemented()
df <- data.frame(lambda=lambda, prior=prior, ll=ll, posterior=post)
df <- data.frame(lambda=lambda, prior=prior, posterior=post)
p <- ggplot(df) + geom_line(aes(lambda, prior, col='prior')) +
  #geom_line(aes(lambda, ll, col='likelihood')) +
  geom_line(aes(lambda, post, col='posterior')) +
  theme_bw()

p
```

In [6]: Grade cell: cell-8db8b258b550abec Score: 13.21 / 13.21 (Top)

Hidden Tests Redacted

Congratulations! All test cases in this cell passed.

1 (c) [4 points] Numerically verify that the posterior is proper by checking that the posterior integrates to 1. Store the result of the integrate function in a variable labeled, `verification`. Note: `integrate()` returns a list. Be sure your variable `verification` is a list resulting from a call to the `integrate` function.

In [7]: Student's answer (Top)

```
# your code here
#.NotYetImplemented()
verification = integrate(posterior_func, 0, Inf)
#verification$val
```

In [8]: Grade cell: cell-7149dbcf2ac7fd20 Score: 7.55 / 7.55 (Top)

Hidden Tests Redacted

Congratulations! All test cases in this cell passed.

Problem 2: From the prior to the posterior predictive distribution¶

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$ with σ^2 known and $\mu \sim N(\mu_0, \sigma_0^2)$. That is, we are working with a normal-normal conjugate prior scenario. If we let σ_0^2 be very large, then we are effectively working with the Jeffreys prior $\pi(\mu) = c$, for any constant c . That is because, as $\sigma_0^2 \rightarrow \infty$, the pdf of a $N(\mu_0, \sigma_0^2)$ converges to a flat line over the real number line.

Define the *prior predictive distribution* as

$$f_{\text{prior}}(x^*) = \int f(x^*|\theta)\pi(\theta)d\theta,$$

where $f(x^*|\theta)$ is the marginal likelihood for a new observation of $X^* = x^*$. One might use this prior predictive distribution to make predictions in the absence of new data. Once new data are observed (as in part (d) below), one should use the *posterior* predictive distribution as described in videos earlier in this course.

Given this definition, one can show that $X^* \sim N(\mu_0, \sigma^2 + \sigma_0^2)$. But, to avoid having to do the integral, let's use Monte Carlo simulations!

2 (a) [5 points] Run a Monte Carlo simulation to verify $X^* \sim N(\mu_0, \sigma^2 + \sigma_0^2)$ numerically. Specifically,

- Generate $N = 100,000$ values of μ from the prior $\pi(\theta) = N(\mu_0 = 10, \sigma_0^2 = 1000)$ (technically a normal prior, but with such a large variance, it is close to the Jeffreys prior). Store these values in a vector labeled `mu`.
- Generate $N = 100,000$ values of x^* from the (marginal) likelihood $f(x^*|\theta) = N(\mu, \sigma^2 = 9)$, where μ are the values in `mu`. Save these values in a vector labeled `xstar`.
- Use these values to calculate the mean and variance of the prior predictive distribution for X^* . Store the mean in a variable labeled `pp_mean` and the variance in a variable labeled `pp_var`. Note: `pp_mean` and `pp_var` should reflect the theoretical result $X^* \sim N(\mu_0, \sigma^2 + \sigma_0^2)$.

In [9]: Student's answer (Top)

```
# your code here
#.NotYetImplemented()
set.seed(0110)
N = 100000 # number of draws from the posterior predictive distribution
mu0 <- 10
sigma0 <- sqrt(1000)
sigma <- 3
mu <- rnorm(N, mu0, sigma0)
xstar <- rnorm(N, mu, sigma)
pp_mean <- mean(xstar)
pp_var <- var(xstar)
pp_mean
pp_var
```

In [10]: Grade cell: cell-20bf85b6ed9e8fa2 Score: 9.43 / 9.43 (Top)

Hidden Tests Redacted

Congratulations! All test cases in this cell passed.

2 (b) [1 point] Using ggplot, plot a histogram of the draws from the prior predictive distribution and save it in a variable labeled `pp_p`.

In [11]: Student's answer (Top)

```
# your code here
#.NotYetImplemented()
pp_p <- ggplot(data.frame(xstar=xstar)) + geom_histogram(aes(xstar))
pp_p
```

In [12]: Grade cell: cell-0d873558be79b01c Score: 1.89 / 1.89 (Top)

Hidden Tests Redacted

Congratulations! All test cases in this cell passed.

2 (c) [3 point] [Multiple Choice] What is the shape of the distribution?

Select the best response to this question and storing the number associated with the response in a variable labeled `choice`.

- The distribution is left skewed with a mode of 10.
- The distribution is bell-shaped (i.e. resembles a normal distribution) centered at 10.
- The distribution is right skewed with a mode of 10.

For example, if you believe the best response is #3, code: `choice=3` in the code block below.

In [13]: Student's answer (Top)

```
# your code here
#.NotYetImplemented()
choice = 2
```

In [14]: Grade cell: cell-6563da9a412bbf97 Score: 5.66 / 5.66 (Top)

Hidden Tests Redacted

